

Divya Ramesh

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EDUCATION

- Columbia University** New York, NY
M.S. in Computer Science (Natural Language Processing) Dec 2026
- Coursework: Data Structures and Algorithms, Machine Learning, Databases, Spoken Language Processing, Computer Networks, Designing LLM Agents, Neural Networks & Deep Learning
- University of California, Los Angeles** Los Angeles, CA
B.A. in Psychology, B.A. in English (GPA: 3.94 / 4.0) Jun 2024
- Coursework: Neural Networks, Cognitive Psychology, Web Development, Statistics
 - Honors: Cum Laude, Dean's Honor List (all quarters), Phi Beta Kappa

SKILLS

Languages: Python, C/C++, R Studio, Java, JavaScript, SQL
Technologies: Colab, RAG, XGBoost, Databricks, HuggingFace, NumPy, Scikit-Learn, Tensorflow, Agentic AI
Skills: Written & Verbal Communication, Data Analysis, Machine Learning, AI Research, Fine Tuning, Feature Engineering, Supervised Learning, Unsupervised Learning, Reinforcement Learning

EXPERIENCE

- Comcast** New York, NY
AI Research Intern June 2026 - August 2026
- Designed and implemented LLM-as-a-Judge evaluation frameworks to systematically assess the output quality and performance of conversational, agentic AI systems
 - Developed a pairwise learning-to-rank pipeline using production logs and human annotations to train a RAG-based Q&A agent
 - Engineered features and designed a supervised learning re-ranking strategy for a document retrieval system to balance broad article relevance with support for narrower, niche search queries
 - Improved retrieval quality by 4% and answer quality by 2%
- Spoken Language Processing Group** New York, NY
Research Assistant Sep 2025 - Present
- Evaluating Large Language Models (LLMs) for multicultural clinical NLP tasks, measuring cultural competency, bias, and performance across culturally specific mental health dialogue datasets
 - Developing LLM evaluation and benchmarking frameworks for cultural competency, model selection, and deployment in therapeutic AI systems, incorporating data analysis and bias assessment

PROJECTS

- Decentralized P2P Payment Application** New York, NY
Python, Blockchain, Socket Programming, Cryptography, Distributed Systems April 2026 - May 2026
- Built a decentralized payment platform on a custom blockchain with proof-of-work mining, enabling users to send and track payments across a P2P network without any central authority
 - Designed a tracker-peer architecture with multi-threaded peers handling concurrent mining, transaction validation, fork resolution, and hash-based signature verification to maintain a consistent distributed ledger
- Speech Emotion Recognition Classifier** New York, NY
scikit-learn, openSMILE, Parselmouth April 2026
- Built a speaker-independent emotion recognition pipeline using the INTERSPEECH 2009 openSMILE feature set (384 features) with per-speaker Z-score normalization across 7 speakers and 15 emotion classes
 - Trained a Random Forest classifier with Leave-One-Group-Out cross-validation, achieving ~4x above-chance accuracy on a challenging 15-class speaker-independent task
 - Analyzed pitch and intensity distributions across emotion categories, finding high-arousal emotions (hot-anger, shame) were most reliably classified with $F1 \geq 0.40$
- Semantic Role Labeling with BERT** New York, NY
PyTorch, HuggingFace Transformers Aug 2025
- Constructed a PropBank-style semantic role labeling system treating SRL as a B-I-O sequence tagging problem; leveraged segment embeddings to encode predicate positions.
 - Fine-tuned BERT for token-level classification with a custom head to predict semantic roles